

No.	Time	Source	Destination	Protocol	Length	Info
10	4.694850	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-1.html HTTP/1.1
12	4.718993	128.119.245.12	192.168.1.102	HTTP	439	HTTP/1.1 200 OK (text/html)
13	4.724332	192.168.1.102	128.119.245.12	HTTP	541	GET /favicon.ico HTTP/1.1
14	4.750366	128.119.245.12	192.168.1.102	HTTP	1395	HTTP/1.1 404 Not Found (text/html)

3. IP address of computer . IP address of server

由於是下載檔案

所以IP為原檔案執行者的

後續在第2題測試中得知我的電腦IP為 192.168.18.158

```

> Frame 10: 555 bytes on wire (4440 bits), 555 bytes captured (4440 bits)
> Ethernet II, Src: Dell_4f:36:23 (00:08:74:4f:36:23), Dst: LinksysGroup_da:af:73 (00:06:25:da:af:73)
> Internet Protocol Version 4, Src: 192.168.1.102, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 4127, Dst Port: 80, Seq: 1, Ack: 1, Len: 501
> Hypertext Transfer Protocol
  > GET /ethereal-labs/lab2-1.html HTTP/1.1\r\n
    Request Method: GET
    Request URI: /ethereal-labs/lab2-1.html
    Request Version: HTTP/1.1
    Host: gaia.cs.umass.edu\r\n
    User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.0.2) Gecko/20021120 Netscape/7.01\r\n
    Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,video/x-mng,i
    Accept-Language: en-us,en;q=0.50\r\n
    Accept-Encoding: gzip, deflate, compress;q=0.9\r\n
    Accept-Charset: ISO-8859-1, utf-8;q=0.66, *;q=0.66\r\n
    Keep-Alive: 300\r\n
    Connection: keep-alive\r\n
    \r\n
    [Response in frame: 12]
    [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-1.html]
  
```

1. browser running HTTP 1.1 #

2. accept language : en-us. English (USA). #

```

0030 fa f0 39 a2 00 00 47 45 54 20 2f 65 74 68 65 72 99 GE T /ether
0040 65 61 6c 2d 6c 61 62 73 2f 6c 61 62 32 2d 31 2e eal-labs /lab2-1.
0050 68 74 6d 6c 20 48 54 54 50 2f 31 2e 31 0d 0a 48 html HTTP/1.1..H
0060 6f 73 74 3a 20 67 61 69 61 2e 63 73 2e 75 6d 61 ost: gai a.cs.uma
0070 73 73 2e 65 64 75 0d 0a 55 73 65 72 2d 41 67 65 ss.edu.. User-Age
0080 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 nt: Mozi lla/5.0
0090 28 57 69 6e 64 6f 77 73 3b 20 55 3b 20 57 69 6e (Windows ; U; Win
00a0 64 6f 77 73 20 4e 54 20 35 2e 31 3b 20 65 6e 2d dows NT 5.1; en-
00b0 55 53 3b 20 72 76 3a 31 2e 30 2e 32 29 20 47 65 US; rv:1 .0.2) Ge
00c0 63 6b 6f 2f 32 30 30 32 31 31 32 30 20 4e 65 74 cko/2002 1120 Net
00d0 73 63 61 70 65 2f 37 2e 30 31 0d 0a 41 63 63 65 scape/7. 01..Acce
00e0 70 74 3a 20 74 65 78 74 2f 78 6d 6c 2c 61 70 70 pt: text /xml,app
00f0 6c 69 63 61 74 69 6f 6e 2f 78 6d 6c 2c 61 70 70 lication /xml,app
0100 6c 69 63 61 74 69 6f 6e 2f 78 68 74 6d 6c 2b 78 lication /xhtml+x
0110 6d 6c 2c 74 65 78 74 2f 68 74 6d 6c 3b 71 3d 30 ml,text/ html;q=0
0120 2e 39 2c 74 65 78 74 2f 70 6c 61 69 6e 3b 71 3d .9,text/ plain;q=
0130 30 2e 38 2c 76 69 64 65 6f 2f 78 2d 6d 6e 67 2c 0.8,vide o/x-mng,
0140 69 6d 61 67 65 2f 70 6e 67 2c 69 6d 61 67 65 2f image/pn g,image/
0150 6a 70 65 67 2c 69 6d 61 67 65 2f 67 69 66 3b 71 jpeg,ima ge/gif;q
0160 3d 30 2e 32 2c 74 65 78 74 2f 63 73 73 2c 2a 2f =0.2,tex t/css,*
0170 2a 3b 71 3d 30 2e 31 0d 0a 41 63 63 65 70 74 2d *;q=0.1. Accept-
0180 4c 61 6e 67 75 61 67 65 3a 20 65 6e 2d 75 73 2c Language : en-us,
  
```

No.	Time	Source	Destination	Protocol	Length	Info
10	4.694850	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-1.html HTTP/1.1
12	4.718993	128.119.245.12	192.168.1.102	HTTP	439	HTTP/1.1 200 OK (text/html)
13	4.724332	192.168.1.102	128.119.245.12	HTTP	541	GET /favicon.ico HTTP/1.1
14	4.750366	128.119.245.12	192.168.1.102	HTTP	1395	HTTP/1.1 404 Not Found (text/html)

- Frame 12: 439 bytes on wire (3512 bits), 439 bytes captured (3512 bits)
- Ethernet II, Src: LinksysGroup_da:af:73 (00:06:25:da:af:73), Dst: Dell_4f:36:23 (00:08:74:4f:36:23)
- Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.1.102
- Transmission Control Protocol, Src Port: 80, Dst Port: 4127, Seq: 1, Ack: 502, Len: 385
- Hypertext Transfer Protocol

HTTP/1.1 200 OK\r\n

Response Version: HTTP/1.1

Status Code: 200

[Status Code Description: OK]

Response Phrase: OK

4: server response 200 OK

Date: Tue, 23 Sep 2003 05:29:50 GMT\r\n

Server: Apache/2.0.40 (Red Hat Linux)\r\n

Last-Modified: Tue, 23 Sep 2003 05:29:00 GMT\r\n

5: file 最后一次在服务器上修改

ETag: "1bfed-49-79d5bf00"\r\n

Accept-Ranges: bytes\r\n

Content-Length: 73\r\n

Keep-Alive: timeout=10, max=100\r\n

Connection: Keep-Alive\r\n

Content-Type: text/html; charset=ISO-8859-1\r\n

\r\n

[Request in frame: 10]

Text item (text), 17 byte(s)

6: 73 bytes of content will be returned.

0030	19 20 7a 1c 00 00 48 54 54 50 2f 31 2e 31 20 32	z...HT TP/1.1 2
0040	30 30 20 4f 4b 0d 0a 44 61 74 65 3a 20 54 75 65	00 OK..D ate: Tue
0050	2c 20 32 33 20 53 65 70 20 32 30 30 33 20 30 35	, 23 Sep 2003 05
0060	3a 32 39 3a 35 30 20 47 4d 54 0d 0a 53 65 72 76	:29:50 G MT..Serv
0070	65 72 3a 20 41 70 61 63 68 65 2f 32 2e 30 2e 34	er: Apac he/2.0.4
0080	30 20 28 52 65 64 20 48 61 74 20 4c 69 6e 75 78	0 (Red H at Linux
0090	29 0d 0a 4c 61 73 74 2d 4d 6f 64 69 66 69 65 64	)..Last- Modified
00a0	3a 20 54 75 65 2c 20 32 33 20 53 65 70 20 32 30	: Tue, 2 3 Sep 20
00b0	30 33 20 30 35 3a 32 39 3a 30 30 20 47 4d 54 0d	03 05:29 :00 GMT=
00c0	0a 45 54 61 67 3a 20 22 31 62 66 65 64 2d 34 39	.ETag: " 1bfed-49
00d0	2d 37 39 64 35 62 66 30 30 22 0d 0a 41 63 63 65	-79d5bf00".Acce
00e0	70 74 2d 52 61 6e 67 65 73 3a 20 62 79 74 65 73	pt-Range s: bytes
00f0	0d 0a 43 6f 6e 74 65 6e 74 2d 4c 65 6e 67 74 68	..Conten t-Length
0100	3a 20 37 33 0d 0a 4b 65 65 70 2d 41 6c 69 76 65	: 73..Ke ep-Alive
0110	3a 20 74 69 6d 65 6f 75 74 3d 31 30 2c 20 6d 61	: timeou t=10, ma
0120	78 3d 31 30 30 0d 0a 43 6f 6e 6e 65 63 74 69 6f	x=100..C onnectio
0130	6e 3a 20 4b 65 65 70 2d 41 6c 69 76 65 0d 0a 43	n: Keep- Alive..C
0140	6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20 74 65 78	ntent-T ype: tex
0150	74 2f 68 74 6d 6c 3b 20 63 68 61 72 73 65 74 3d	t/html; charset=
0160	49 53 4f 2d 38 38 35 39 2d 31 0d 0a 0d 0a 3c 68	ISO-8859 -1....<h
0170	74 6d 6c 3e 0a 43 6f 6e 67 72 61 74 75 6c 61 74	tml> Con gratulat
0180	69 6f 6e 73 2e 20 20 59 6f 75 27 76 65 20 64 6f	ions. Y ou've do

自行測試



http

No.	Time	Source	Destination	Protocol	Length	Info
146	7.753345	192.168.239.158	128.119.245.12	HTTP	559	GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
152	7.992522	128.119.245.12	192.168.239.158	HTTP	540	HTTP/1.1 200 OK (text/html)

3 = my computer ID address . server ID address

1. client running HTTP 1.1

```

Frame 146: 559 bytes on wire (4472 bits), 559 bytes captured (4472 bits) on interface \Device\NPF_{3EC0E...
Ethernet II, Src: AzureWaveTec_2a:81:41 (ec:2e:98:2a:81:41), Dst: 86:34:07:42:56:38 (86:34:07:42:56:38)
Internet Protocol Version 4, Src: 192.168.239.158, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 51984, Dst Port: 80, Seq: 1, Ack: 1, Len: 505
Hypertext Transfer Protocol
  GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1\r\n
    Request Method: GET
    Request URI: /wireshark-labs/HTTP-wireshark-file1.html
    Request Version: HTTP/1.1
    Host: gaia.cs.umass.edu\r\n
    Connection: keep-alive\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/1...
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=...
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: zh-TW,zh;q=0.9,en-US;q=0.8,en;q=0.7,zh-CN;q=0.6\r\n
  \r\n
  [Response in frame: 152]
  [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file1.html]
    
```

```

0000 86 34 07 42 56 38 ec 2e 98 2a 81 41 08 00 45 00 -4 BV8... * A-E
0010 02 21 1e 37 40 00 80 06 b4 d4 c0 a8 ef 9e 80 77 -1 7@... ..w
0020 f5 0c cb 10 00 50 91 5e b6 8e 78 4c 3b a7 50 18 -... P^...xL:P
0030 02 01 26 17 00 00 47 45 54 20 2f 77 69 72 65 73 --&...GE T/wires
0040 68 61 72 6b 2d 6c 61 62 73 2f 48 54 54 50 2d 77 hark-lab s/HTTP-w
0050 69 72 65 73 68 61 72 6b 2d 66 69 6c 65 31 2e 68 ireshark -file1.h
0060 74 6d 6c 20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f tml HTTP /1.1--Ho
0070 73 74 3a 20 67 61 69 61 2e 63 73 2e 75 6d 61 73 st: gaia .cs.umass
0080 73 2e 65 64 75 0d 0a 43 6f 6e 6e 65 63 74 69 6f s.edu : C connectio
0090 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65 0d 0a 55 n: keep- alive- U
00a0 70 67 72 61 64 65 2d 49 6e 73 65 63 75 72 65 2d pgrade- I nsecure-
00b0 52 65 71 75 65 73 74 73 3a 20 31 0d 0a 55 73 65 Requests : 1- Use
00c0 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 r-Agent: Mozilla
00d0 2f 35 2e 30 20 28 57 69 6e 64 6f 77 73 20 4e 54 /5.0 (Wi ndows NT
00e0 20 31 30 2e 30 3b 20 57 69 6e 36 34 3b 20 78 36 10.0; W in64; x6
00f0 34 29 20 41 70 70 6c 65 57 65 62 4b 69 74 2f 35 4) Apple WebKit/5
0100 33 37 2e 33 36 20 28 4b 48 54 4d 4c 2c 20 6c 69 37.36 (K HTML, li
0110 6b 65 20 47 65 63 6b 6f 29 20 43 68 72 6f 6d 65 ke Gecko ) Chrome
0120 2f 31 33 30 2e 30 2e 30 2e 30 20 53 61 66 61 72 /130.0.0 .0 Safari
0130 69 2f 35 33 37 2e 33 36 0d 0a 41 63 63 65 70 74 i/537.36 -Accept
0140 3a 20 74 65 78 74 2f 68 74 6d 6c 2c 61 70 70 6c : text/h tml,appl
0150 69 63 61 74 69 6f 6e 2f 78 68 74 6d 6c 2b 78 6d ication/ xhtml+xm
    
```



http

No.	Time	Source	Destination	Protocol	Length	Info
146	7.753345	192.168.239.158	128.119.245.12	HTTP	559	GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
152	7.992522	128.119.245.12	192.168.239.158	HTTP	540	HTTP/1.1 200 OK (text/html)

Frame 152: 540 bytes on wire (4320 bits), 540 bytes captured (4320 bits) on interface \Device\NPF_{3EC...}

Ethernet II, Src: 86:34:07:42:56:38 (86:34:07:42:56:38), Dst: AzureWaveTec_2a:81:41 (ec:2e:98:2a:81:41)

Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.239.158

Transmission Control Protocol, Src Port: 80, Dst Port: 51984, Seq: 1, Ack: 506, Len: 486

Hypertext Transfer Protocol

HTTP/1.1 200 OK\r\n

Response Version: HTTP/1.1

Status Code: 200

[Status Code Description: OK]

Response Phrase: OK

Date: Thu, 31 Oct 2024 20:20:05 GMT\r\n

Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.33 mod_perl/2.0.11 Perl/v5.16.3\r\n

Last-Modified: Thu, 31 Oct 2024 05:59:02 GMT\r\n

ETag: "80-625bf83e59b04"\r\n

Accept-Ranges: bytes\r\n

Content-Length: 128\r\n

Keep-Alive: timeout=5, max=100\r\n

Connection: Keep-Alive\r\n

Content-Type: text/html; charset=UTF-8\r\n\r\n

4 response 200 ok #

5. 2024.10.31. 5:59

6: return 128 bytes #

```

0000  ec 2e 98 2a 81 41 86 34 07 42 56 38 08 00 45 00  ...* A.4 -BV8--E
0010  02 0e 40 3f 40 00 21 06 f1 df 80 77 f5 0c c0 a8  ...@?@ !...w....
0020  ef 9e 00 50 cb 10 78 4c 3b a7 91 5e b8 87 50 18  ...P...xL ;...^...P
0030  00 ed 21 14 00 00 48 54 54 50 2f 31 2e 31 20 32  ...!...HT TP/1.1 2
0040  30 30 20 4f 4b 0d 0a 44 61 74 65 3a 20 54 68 75  00 OK...D ate: Thu
0050  2c 20 33 31 20 4f 63 74 20 32 30 32 34 20 32 30  , 31 Oct 2024 20
0060  3a 32 30 3a 30 35 20 47 4d 54 0d 0a 53 65 72 76  :20:05 G MT . Serv
0070  65 72 3a 20 41 70 61 63 68 65 2f 32 2e 34 2e 36  er: Apac he/2.4.6
0080  20 28 43 65 6e 74 4f 53 29 20 4f 70 65 6e 53 53  (CentOS ) OpenSS
0090  4c 2f 31 2e 30 2e 32 6b 2d 66 69 70 73 20 50 48  L/1.0.2k -fips PH
00a0  50 2f 37 2e 34 2e 33 33 20 6d 6f 64 5f 70 65 72  P/7.4.33 mod_per
00b0  6c 2f 32 2e 30 2e 31 31 20 50 65 72 6c 2f 76 35  l/2.0.11 Perl/v5
00c0  2e 31 36 2e 33 0d 0a 4c 61 73 74 2d 4d 6f 64 69  .16.3 L ast-Modi
00d0  66 69 65 64 3a 20 54 68 75 2c 20 33 31 20 4f 63  fied: Thu, 31 Oc
00e0  74 20 32 30 32 34 20 30 35 3a 35 39 3a 30 32 20  t 2024 0 5:59:02
00f0  47 4d 54 0d 0a 45 54 61 67 3a 20 22 38 30 2d 36  GMT: ETa g: "80-6
0100  32 35 62 66 38 33 65 35 39 62 30 34 22 0d 0a 41  25bf83e5 9b04" . A
0110  63 63 65 70 74 2d 52 61 6e 67 65 73 3a 20 62 79  ccept-Ra nges: by
0120  74 65 73 0d 0a 43 6f 6e 74 65 6e 74 2d 4c 65 6e  tes . Con tent-Len
0130  67 74 68 3a 20 31 32 38 0d 0a 4b 65 65 70 2d 41  gth: 128 . Keep-A
0140  6c 69 76 65 3a 20 74 69 6d 65 6f 75 74 3d 35 2c  live: ti meout=5,
0150  20 6d 61 78 3d 31 30 30 0d 0a 43 6f 6e 6e 65 63  max=100 .Connec

```

7: There is no any headers within the data that are not displayed. #

```
▼ Hypertext Transfer Protocol
  ▶ GET /ethereal-labs/lab2-1.html HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.0.2) Gecko/20021120 Netscape/7.01\r\n
    Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,video/x-mng\r\n
    Accept-Language: en-us,en;q=0.5\r\n
    Accept-Encoding: gzip, deflate, compress;q=0.9\r\n
    Accept-Charset: ISO-8859-1, utf-8;q=0.66,*;q=0.66\r\n
    Keep-Alive: 300\r\n
    Connection: keep-alive\r\n
  \r\n
  [Response in frame: 12]
  [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-1.html]
```

header 都有顯示.

```
▼ Hypertext Transfer Protocol
  ▶ HTTP/1.1 200 OK\r\n
    Date: Tue, 23 Sep 2003 05:29:50 GMT\r\n
    Server: Apache/2.0.40 (Red Hat Linux)\r\n
    Last-Modified: Tue, 23 Sep 2003 05:29:00 GMT\r\n
    ETag: "1bfed-49-79d5bf00"\r\n
    Accept-Ranges: bytes\r\n
    Content-Length: 73\r\n
    Keep-Alive: timeout=10, max=100\r\n
    Connection: Keep-Alive\r\n
    Content-Type: text/html; charset=ISO-8859-1\r\n
  \r\n
  [Request in frame: 10]
  [Time since request: 0.024143000 seconds]
  [Request URI: /ethereal-labs/lab2-1.html]
  [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-1.html]
  File Data: 73 bytes
▼ Line-based text data: text/html (3 lines)
  <html>\r\n
  Congratulations. You've downloaded the file lab2-1.html!\r\n
  </html>\r\n
```

該次作業: 1. HTTP 1.1

2. zh-TW (繁體中文, 臺灣).

3. 192.168.18.158 / 128.119.245.12

4. 200 OK

5. 2004.10.31.

6. 128 bytes.

7. 全都有顯示

#

http

No.	Time	Source	Destination	Protocol	Length	Info
113	0.390210	192.168.18.158	128.119.245.12	HTTP	559	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
130	0.618377	128.119.245.12	192.168.18.158	HTTP	784	HTTP/1.1 200 OK (text/html)
161	1.165992	192.168.18.158	128.119.245.12	HTTP	505	GET /favicon.ico HTTP/1.1
185	1.392841	128.119.245.12	192.168.18.158	HTTP	538	HTTP/1.1 404 Not Found (text/html)
670	20.324715	192.168.18.158	128.119.245.12	HTTP	671	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
676	20.554532	128.119.245.12	192.168.18.158	HTTP	294	HTTP/1.1 304 Not Modified

→ IP address of my computer.

Wireshark - Packet 113 - Wi-Fi

Hypertext Transfer Protocol

- GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1\r\n
 Request Method: GET
 Request URI: /wireshark-labs/HTTP-wireshark-file2.html
 Request Version: HTTP/1.1
 Host: gaia.cs.umass.edu\r\n
 Connection: keep-alive\r\n
 Upgrade-Insecure-Requests: 1\r\n
 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/130.0.0.0 Safari/537.36\r\n
 Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n
 Accept-Encoding: gzip, deflate\r\n
 Accept-Language: zh-TW,zh;q=0.9,en-US;q=0.8,en;q=0.7,zh-CN;q=0.6\r\n
 \r\n
 [Response in frame: 130]
 [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html]

8: I don't see an "IF-MODIFIED-SINCE" line in HTTP Get

```

0000  6a e1 15 74 02 43 ec 2e 98 2a 81 41 08 00 45 00  j . t . C . . * . A . E .
0010  02 21 bd 7b 40 00 80 06 f2 90 c0 a8 12 9e 80 77  ! ! { @ . . . . . w
0020  f5 0c c4 3c 00 50 63 8a 94 59 3b 4f 49 b7 50 18  < P c . . Y ; O I P .
0030  02 01 88 e1 00 00 47 45 54 20 2f 77 69 72 65 73  < . . . . . G E T / w i r e s
0040  68 61 72 6b 2d 6c 61 62 73 2f 48 54 54 50 2d 77  h a r k - l a b s / H T T P - w
0050  69 72 65 73 68 61 72 6b 2d 66 69 6c 65 32 2e 68  i r e s h a r k - f i l e 2 . h
0060  74 6d 6c 20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f  t m l H T T P / 1 . 1 . H o
0070  73 74 3a 20 67 61 69 61 2e 63 73 2e 75 6d 61 73  s t : g a i a . c s . u m a s
0080  73 2e 65 64 75 0d 0a 43 6f 6e 6e 65 63 74 69 6f  s . e d u . C o n n e c t i o
0090  6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65 0d 0a 55  n : k e e p - a l i v e . U
00a0  70 67 72 61 64 65 2d 49 6e 73 65 63 75 72 65 2d  p g r a d e - I n s e c u r e -
00b0  52 65 71 75 65 73 74 73 3a 20 31 0d 0a 55 73 65  R e q u e s t s : 1 . U s e
00c0  72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c 61  r - A g e n t : M o z i l l a
00d0  2f 35 2e 30 20 28 57 69 6e 64 6f 77 73 20 4e 54  / 5 . 0 ( W i n d o w s N T
00e0  20 31 30 2e 30 3b 20 57 69 6e 36 34 3b 20 78 36  1 0 . 0 ; W i n 6 4 ; x 6
00f0  34 29 20 41 70 70 6c 65 57 65 62 4b 69 74 2f 35  4 ) A p p l e W e b K i t / 5
0100  33 37 2e 33 36 20 28 4b 48 54 4d 4c 2c 20 6c 69  3 7 . 3 6 ( K H T M L , l i
0110  6b 65 20 47 65 63 6b 6f 29 20 43 68 72 6f 6d 65  k e G e c k o ) C h r o m e

```

No.	Time	Source	Destination	Protocol	Length	Info
113	0.390210	192.168.18.158	128.119.245.12	HTTP	559	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
130	0.618377	128.119.245.12	192.168.18.158	HTTP	784	HTTP/1.1 200 OK (text/html)
161	1.165992	192.168.18.158	128.119.245.12	HTTP	505	GET /favicon.ico HTTP/1.1
185	1.392841	128.119.245.12	192.168.18.158	HTTP	538	HTTP/1.1 404 Not Found
670	20.324715	192.168.18.158	128.119.245.12	HTTP	671	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
676	20.554532	128.119.245.12	192.168.18.158	HTTP	294	HTTP/1.1 304 Not Modified

9. 200 OK 代表 request 成功, server 回傳 file
加上, 下方也顯示出了 content

```
Keep-Alive: timeout=5, max=100\r\n
Connection: Keep-Alive\r\n
Content-Type: text/html; charset=UTF-8\r\n
\r\n
```

[Request in frame: 113]

[Time since request: 0.228167000 seconds]

[Request URI: /wireshark-labs/HTTP-wireshark-file2.html]

[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html]

File Data: 371 bytes

Line-based text data: text/html (10 lines)

```
\n
<html>\n
\n
Congratulations again! Now you've downloaded the file lab2-2.html. <br>\n
This file's last modification date will not change. <p>\n
Thus if you download this multiple times on your browser, a complete copy <br>\n
will only be sent once by the server due to the inclusion of the IN-MODIFIED-SINCE<br>\n
field in your browser's HTTP GET request to the server.\n
\n
</html>\n
```

Wireshark - Packet 130 - Wi-Fi

- Frame 130: 784 bytes on wire (6272 bits), 784 bytes captured (6272 bits) on interface \Device\NPF...
- Ethernet II, Src: 6a:e1:15:74:02:43 (6a:e1:15:74:02:43), Dst: AzureWaveTec_2a:81:41 (ec:2e:98:2a:...
- Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.18.158
- Transmission Control Protocol, Src Port: 80, Dst Port: 50236, Seq: 1, Ack: 506, Len: 730

Hypertext Transfer Protocol

HTTP/1.1 200 OK\r\n

Response Version: HTTP/1.1

Status Code: 200

[Status Code Description: OK]

Response Phrase: OK

Date: Thu, 31 Oct 2024 15:56:38 GMT\r\n

Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.33 mod_perl/2.0.11 Perl/v5.16.3\r\n

Last-Modified: Thu, 31 Oct 2024 05:59:02 GMT\r\n

ETag: "173-625bf83e59334"\r\n

Accept-Ranges: bytes\r\n

```
0000 ec 2e 98 2a 81 41 6a e1 15 74 02 43 08 00 45 00 ...*Aj...t.C...E
0010 03 02 01 bc 40 00 20 06 0d 70 80 77 f5 0c c0 a8 ...@...p.w...
0020 12 9e 00 50 c4 3c 3b 4f 49 b7 63 8a 96 52 50 18 ...P<;O I c...RP
0030 00 ed c4 ce 00 00 48 54 54 50 2f 31 2e 31 20 32 ...HT TP/1.1 2
0040 30 30 20 4f 4b 0d 0a 44 61 74 65 3a 20 54 68 75 ...OK OK ate: Thu
0050 2c 20 33 31 20 4f 63 74 20 32 30 32 34 20 31 35 ...31 Oct 2024 15
0060 3a 35 36 3a 33 38 20 47 4d 54 0d 0a 53 65 72 76 ...:56:38 G MT...Serv
0070 65 72 3a 20 41 70 61 63 68 65 2f 32 2e 34 2e 36 ...er: Apac he/2.4.6
0080 20 28 43 65 6e 74 4f 53 29 20 4f 70 65 6e 53 53 ... (CentOS ) OpenSS
0090 4c 2f 31 2e 30 2e 32 6b 2d 66 69 70 73 20 50 48 ...L/1.0.2k -fips PH
00a0 50 2f 37 2e 34 2e 33 33 20 6d 6f 64 5f 70 65 72 ...P/7.4.33 mod_per
00b0 6c 2f 32 2e 30 2e 31 31 20 50 65 72 6c 2f 76 35 ...l/2.0.11 Perl/v5
00c0 2e 31 36 2e 33 0d 0a 4c 61 73 74 2d 4d 6f 64 69 ...16.3..L ast-Modi
00d0 66 69 65 64 3a 20 54 68 75 2c 20 33 31 20 4f 63 ...fied: Th u, 31 Oc
00e0 74 20 32 30 32 34 20 30 35 3a 35 39 3a 30 32 20 ...t 2024 0 5:59:02
00f0 47 4d 54 0d 0a 45 54 61 67 3a 20 22 31 37 33 2d ...GMT -ETa g: "173-
0100 36 32 35 62 66 38 33 65 35 39 33 33 34 22 0d 0a ...625bf83e 59334"...
0110 41 63 63 65 70 74 2d 52 61 6e 67 65 73 3a 20 62 ...Accept-R anges: b
```

No.: 130 - Time: 0.618377 - Source: 128.119.245.12 - Destination: 192.168.18.158 - Protocol: HTTP - Length: 784 - Info: HTTP/1.1 200 OK (text/html)

Show packet bytes

Layout: Vertical (Stacked)

http

No.	Time	Source	Destination	Protocol	Length	Info
113	0.390210	192.168.18.158	128.119.245.12			
130	0.618377	128.119.245.12	192.168.18.158			
161	1.165992	192.168.18.158	128.119.245.12			
185	1.392841	128.119.245.12	192.168.18.158			
670	20.324715	192.168.18.158	128.119.245.12			
676	20.554532	128.119.245.12	192.168.18.158			

Wireshark · Packet 670 · Wi-Fi

Request URI: /wireshark-labs/HTTP-wireshark-file2.html

Request Version: HTTP/1.1

Host: gaia.cs.umass.edu\r\n

Connection: keep-alive\r\n

Cache-Control: max-age=0\r\n

Upgrade-Insecure-Requests: 1\r\n

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/130.0.0.0 Safari/537.36\r\n

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.1\r\n

Accept-Encoding: gzip, deflate\r\n

Accept-Language: zh-TW,zh;q=0.9,en-US;q=0.8,en;q=0.7,zh-CN;q=0.6\r\n

If-None-Match: "173-625bf83e59334"\r\n

If-Modified-Since: Thu, 31 Oct 2024 05:59:02 GMT\r\n

\r\n

[Response in frame: 676]

[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html]

10. 出現 "IF-MODIFIED-SINCE" line
And the header information: last modified date ✖

No.	Time	Source	Destination	Protocol	Length	Info
113	0.390210	192.168.18.158	128.119.245.12	HTTP	559	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
130	0.618377	128.119.245.12	192.168.18.158	HTTP	784	HTTP/1.1 200 OK (text/html)
161	1.165992	192.168.18.158	128.119.245.12	HTTP	505	GET /favicon.ico HTTP/1.1
185	1.392841	128.119.245.12	192.168.18.158	HTTP	538	HTTP/1.1 404 Not Found (text/html)
670	20.324715	192.168.18.158	128.119.245.12	HTTP	671	GET /wireshark-lab
676	20.554532	128.119.245.12	192.168.18.158	HTTP	294	HTTP/1.1 304 Not Modified

11. The server response 304 Not Modified

304 Not Modified indicates that the requested source hasn't been modified since the last time it was loaded, and there is no need to transfer it again. 因此才要叫我們清除快取，由於快取中有上次 load 的版本，server 認為無需再傳一次。

```

Frame 676: 294 bytes on wire (2352 bits), 294 bytes captured (2352 bits) on interface \Device\NPF...
Ethernet II, Src: 6a:e1:15:74:02:43 (6a:e1:15:74:02:43), Dst: AzureWaveTec_2a:81:41 (ec:2e:98:2a:81:41)
Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.18.158
Transmission Control Protocol, Src Port: 80, Dst Port: 50260, Seq: 1, Ack: 618, Len: 240
Hypertext Transfer Protocol
  HTTP/1.1 304 Not Modified\r\n
    Response Version: HTTP/1.1
    Status Code: 304
    [Status Code Description: Not Modified]
    Response Phrase: Not Modified
    Date: Thu, 31 Oct 2024 15:56:58 GMT\r\n
    Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.33 mod_perl/2.0.11 Perl/v5.16.3\r\n
    Connection: Keep-Alive\r\n
    Keep-Alive: timeout=5, max=100\r\n
    ETag: "173-625bf83e59334"\r\n
    \r\n
[Request in frame: 670]
[Time since request: 0.229817000 seconds]
[Request URI: /wireshark-labs/HTTP-wireshark-file2.html]
[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html]

```

Wireshark - Packet 676 - Wi-Fi

```

Frame 676: 294 bytes on wire (2352 bits), 294 bytes captured (2352 bits) on interface \Device\NPF...
Ethernet II, Src: 6a:e1:15:74:02:43 (6a:e1:15:74:02:43), Dst: AzureWaveTec_2a:81:41 (ec:2e:98:2a:81:41)
Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.18.158
Transmission Control Protocol, Src Port: 80, Dst Port: 50260, Seq: 1, Ack: 618, Len: 240
Hypertext Transfer Protocol
  HTTP/1.1 304 Not Modified\r\n
    Response Version: HTTP/1.1
    Status Code: 304
    [Status Code Description: Not Modified]
    Response Phrase: Not Modified
    Date: Thu, 31 Oct 2024 15:56:58 GMT\r\n
    Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.33 mod_perl/2.0.11 Perl/v5.16.3\r\n
    Connection: Keep-Alive\r\n

```

```

0000 ec 2e 98 2a 81 41 6a e1 15 74 02 43 08 00 45 00 ...*Aj -t C-E
0010 01 18 53 48 40 00 20 06 bd cd 80 77 f5 0c c0 a8 ...SH@...w...
0020 12 9e 00 50 c4 54 46 e3 c6 a9 4a 03 b9 e3 50 18 ...P.TF...J...P
0030 00 ee db 60 00 00 48 54 54 50 2f 31 2e 31 20 33 ...-HT TP/1.1 3
0040 30 34 20 4e 6f 74 20 4d 6f 64 69 66 69 65 64 0d 04 Not Modified
0050 0a 44 61 74 65 3a 20 54 68 75 2c 20 33 31 20 4f -Date: T hu, 31 O
0060 63 74 20 32 30 32 34 20 31 35 3a 35 36 3a 35 38 ct 2024 15:56:58
0070 20 47 4d 54 0d 0a 53 65 72 76 65 72 3a 20 41 70 GMT Se rver: Ap
0080 61 63 68 65 2f 32 2e 34 2e 36 20 28 43 65 6e 74 ache/2.4 .6 (Cent
0090 4f 53 29 20 4f 70 65 6e 53 53 4c 2f 31 2e 30 2e OS) Open SSL/1.0.
00a0 32 6b 2d 66 69 70 73 20 50 48 50 2f 37 2e 34 2e 2k-fips PHP/7.4.
00b0 33 33 20 6d 6f 64 5f 70 65 72 6c 2f 32 2e 30 2e 33 mod_p erl/2.0.
00c0 31 31 20 50 65 72 6c 2f 76 35 2e 31 36 2e 33 0d 11 Perl/ v5.16.3
00d0 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20 4b 65 65 -Connect ion: Kee
00e0 70 2d 41 6c 69 76 65 0d 0a 4b 65 65 70 2d 41 6c p-Alive -Keep-Al
00f0 69 76 65 3a 20 74 69 6d 65 6f 75 74 3d 35 2c 20 ive: tim eout=5,

```

No.: 676 - Time: 20.554532 - Source: 128.119.245.12 - Destination: 192.168.18.158 - Protocol: HTTP - Length: 294 - Info: HTTP/1.1 304 Not Modified

Show packet bytes Layout: Vertical (Stacked)

http-ethereal-trace-3

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
8	4.623732	192.168.1.102	128.119.245.12	HTTP	555	GET /etherreal-labs/lab2-3.html HTTP/1.1
14	4.680920	128.119.245.12	192.168.1.102	HTTP	490	HTTP/1.1 200 OK (text/html)

3: packet 14 has the status code and phrase associated with the response to the HTTP GET request

12: only for 1 HTTP Get request messages and it's packet number is 8.

14: The status code and phrase is 200 OK

Wireshark - Packet 14 - http-ethereal-trace-3

Line-based text data: text/html (98 lines)

```
<html><head> \n
<title>Historical Documents:THE BILL OF RIGHTS</title></head>\n
\n
\n
<body bgcolor="#ffffff" link="#330000" vlink="#666633">\n
<p><br>\n
</p>\n
<p><p><center><b>THE BILL OF RIGHTS</b><br>\n
    <em>Amendments 1-10 of the Constitution</em>\n
</center>\n
\n
<p>The Conventions of a number of the States having, at the time of adopting\n
the Constitution, expressed a desire, in order to prevent misconstruction\n
or abuse of its powers, that further declaratory and restrictive clauses\n
```

Wireshark - Packet 8 - http-ethereal-trace-3

Frame 8: 555 bytes on wire (4440 bits), 555 bytes captured (4440 bits)

Ethernet II, Src: Dell_4f:36:23 (00:08:74:4f:36:23), Dst: LinksysGroup_da:af:73 (00:06:25:da:af:73)

Internet Protocol Version 4, Src: 192.168.1.102, Dst: 128.119.245.12

Transmission Control Protocol, Src Port: 4272, Dst Port: 80, Seq: 1, Ack: 1, Len: 501

Hypertext Transfer Protocol

- GET /etherreal-labs/lab2-3.html HTTP/1.1\r\n
- Request Method: GET
- Request URI: /etherreal-labs/lab2-3.html
- Request Version: HTTP/1.1
- Host: gaia.cs.umass.edu\r\n
- User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.0.2) Gecko/20021120 Netscape/7.01\r\n
- Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,video/x-mng,...
- Accept-Language: en-us, en;q=0.50\r\n
- Accept-Encoding: gzip, deflate, compress;q=0.9\r\n
- Accept-Charset: ISO-8859-1, utf-8;q=0.66, *;q=0.66\r\n

0000 00 08 74 4f 36 23 00 06 25 da af 73 08 00 45 00 ...t06#...%s.E

0010 01 dc 21 71 40 00 37 06 e9 18 80 77 f5 0c c0 a8 ...lq0 7...w...

0020 01 66 00 50 10 b0 85 b2 bb 80 fb 98 e0 df 50 18 ...f P...P...

0030 19 20 25 ab 00 00 3e 3c 68 33 3e 41 6d 65 6e 64 ...%...> h3>Amend

0040 6d 65 6e 74 20 49 58 3c 2f 68 33 3e 3c 2f 73 74 ...ment IX< /h3><st

0050 72 6f 6e 67 3e 3c 2f 61 3e 0a 0a 3c 70 3e 3c 2f ...rong>...<p></

0060 70 3e 3c 70 3e 54 68 65 20 65 6e 75 6d 65 72 61 ...p>>The enumera

0070 74 69 6f 6e 20 69 6e 20 74 68 65 20 43 6f 6e 73 ...tion in the Cons

0080 74 69 74 75 74 69 6f 6e 2c 20 6f 66 20 63 65 72 ...titution , of cer

0090 74 61 69 6e 20 72 69 67 68 74 73 2c 20 73 68 61 ...tain rig hts, sha

00a0 6c 6c 0a 6e 6f 74 20 62 65 20 63 6f 6e 73 74 72 ...ll-not be constr

00b0 75 65 64 20 74 6f 20 64 65 6e 79 20 6f 72 20 64 ...ued to deny or d

00c0 69 73 70 61 72 61 67 65 20 6f 74 68 65 72 73 20 ...isparage others

00d0 72 65 74 61 69 6e 65 64 20 62 79 20 74 68 65 20 ...retained by the

00e0 70 65 6f 70 6e 65 2e 0a 0a 3c 2f 70 3e 3c 70 3e ...people.. </p><p>

00f0 3c 61 20 6e 61 6d 65 3d 22 31 30 22 3e 3c 73 74 ...<st

0000 00 06 25 da af 73 00 08 74 4f 36 23 00 00 45 00 ...%s...t06#...E

0010 02 1d 02 84 40 00 00 06 00 00 c0 a8 01 66 80 77 ...@...f.w

0020 f5 0c 10 b0 00 50 fb 98 de ea 85 b2 aa 64 50 18 ...P...dP...

0030 fa f0 39 a2 00 00 47 45 54 20 2f 65 74 68 65 72 ...9...GE T /ether

0040 65 61 6c 2d 6c 61 62 73 2f 6c 61 62 32 2d 33 2e ...eal-labs /lab2-3.

0050 68 74 6d 6c 20 48 54 54 50 2f 31 2e 31 0d 0a 48 ...html HTTP/1.1 H

0060 6f 73 74 3a 20 67 61 69 61 2e 63 73 2e 75 6d 61 ...ost: gai a.cs.uma

0070 73 73 2e 65 64 75 0d 0a 55 73 65 72 2d 41 67 65 ...ss.edu... User-Age

0080 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 ...nt: Mozi lla/5.0

0090 28 57 69 6e 64 6f 77 73 3b 20 55 3b 20 57 69 6e ... (Windows ; U; Win

00a0 64 6f 77 73 20 4e 54 20 35 2e 31 3b 20 65 6e 2d ...dows NT 5.1; en-

00b0 55 53 3b 20 72 76 3a 31 2e 30 2e 32 29 20 47 65 ...US; rv:1 .0.2) Ge

00c0 65 6b 6f 2f 32 30 30 32 31 31 32 30 20 4e 65 74 ...cko/2002 1120 Net

00d0 73 63 61 70 65 2f 37 2e 30 31 0d 0a 41 63 63 65 ...scape/7. 01 Acce

00e0 70 74 3a 20 74 65 78 74 2f 78 6d 6c 2c 61 70 70 ...pt: text /xml,app

00f0 6c 69 63 61 74 69 6f 6e 2f 78 6d 6c 2c 61 70 70 ...lication /xml,app

0100 6c 69 63 61 74 69 6f 6e 2f 78 68 74 6d 6c 2b 78 ...lication /html+x

0110 6d 6c 2c 74 65 78 74 2f 68 74 6d 6c 3b 71 3d 30 ...ml;text/ html;q=0

自行測試:

參考檔案

No.	Time	Source	Destination	Protocol	Length	Info
108	6.298079	192.168.18.158	128.119.245.12	HTTP	559	GET /wireshark-labs/HTTP-wireshark- file4.html HTTP/1.1
123	6.527731	128.119.245.12	192.168.18.158	HTTP	1355	HTTP/1.1 200 OK (text/html)
125	6.542688	192.168.18.158	128.119.245.12	HTTP	505	GET /pearson.png HTTP/1.1
138	7.180596	192.168.18.158	178.79.137.164	HTTP	472	GET /BE cover small.jpg HTTP/1.1
142	7.585290	178.79.137.164	192.168.18.158	HTTP	225	HTTP/1.1 301 Moved permanently

1b: In both diagram browser sends 3 HTTP GET request.

first packet: send to 128.119.245.12.

second packet: send to 128.119.245.12

third packet: send to 178.79.137.164

```

Frame 108: 559 bytes on wire (4472 bits), 559 bytes captured (4472 bits) on interface Device\NPF {3ECB...
Ethernet II, Src: AzureWaveTec_2a:81:41 (ac:2e:98:2a:81:41), Dst: 6a:el:15:74:02:43 (6a:el:15:74:02:43)
Internet Protocol Version 4, Src: 192.168.18.158, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 64372, Dst Port: 80, Seq: 1, Ack: 1, Len: 505
Hypertext Transfer Protocol
  GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1\r\n
Host: gaia.cs.umass.edu\r\n
Connection: keep-alive\r\n
Upgrade-Insecure-Requests: 1\r\n
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/1...
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=...
Accept-Encoding: gzip, deflate\r\n
Accept-Language: zh-TW,zh;q=0.9,en-US;q=0.8,en;q=0.7,zh-CN;q=0.6\r\n
\r\n
[Response in frame: 123]
[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file4.html]
    
```

No.	Time	Source	Destination	Protocol	Length	Info
10	7.236929	192.168.1.102	128.119.245.12	HTTP	555	GET /etheral-labs/lab2-4.html HTTP/1.1
12	7.260813	128.119.245.12	192.168.1.102	HTTP	1057	HTTP/1.1 200 OK (text/html)
17	7.305485	192.168.1.102	165.193.123.218	HTTP	625	GET /catalog/images/pearson-logo-footer.gif HTTP/1.1
20	7.308801	192.168.1.102	134.241.6.82	HTTP	609	GET /-kurose/cove.jpg HTTP/1.1
25	7.333054	165.193.123.218	192.168.1.102	HTTP	912	HTTP/1.1 200 OK (image)
54	7.580877	134.241.6.82	192.168.1.102	HTTP	1096	HTTP/1.0 200 Document Follows (JPEG JFIF image)

```

Frame 10: 555 bytes on wire (4440 bits), 555 bytes captured (4440 bits)
Ethernet II, Src: Dell_4f:36:73 (00:00:7d:4f:36:73), Dst: LinksysGroup_da:af:73 (00:06:25:da:af:73)
Internet Protocol Version 4, Src: 192.168.1.102, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 4907, Dst Port: 80, Seq: 1, Ack: 1, Len: 501
Hypertext Transfer Protocol
  GET /etheral-labs/lab2-4.html HTTP/1.1\r\n
Request Method: GET
Request URI: /etheral-labs/lab2-4.html
Request Version: HTTP/1.1
Host: gaia.cs.umass.edu\r\n
User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.0.2) Gecko/20021120 Netscape/7.0.1...
Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,video/*;q=0.8...
Accept-Language: en-us, en;q=0.50\r\n
Accept-Encoding: gzip, deflate, compress;q=0.9\r\n
Accept-Charset: ISO-8859-1, utf-8;q=0.66, *q=0.66\r\n
Keep-Alive: 300\r\n
Connection: keep-alive\r\n
\r\n
[Response in frame: 12]
[Full request URI: http://gaia.cs.umass.edu/etheral-labs/lab2-4.html]
    
```

165.193.123.218 參考檔案

134.241.6.82 參考檔案

17: Two images were downloaded in parallel.

從自行測試結果來看， packet 125 和 138 發送影像請求

但第一張影像 server 未有回應，第二張則是物件位置已更改。

從參考檔案結果來看， packet 17 和 20 發送了影像請求

而 packet 25 和 54 則是 server 回傳影像。



No.	Time	Source	Destination	Protocol	Length	Info
5	3.193022	192.168.18.158	128.119.245.12	HTTP	575	GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1
52	3.460543	128.119.245.12	192.168.18.158	HTTP	771	HTTP/1.1 401 Unauthorized (text/html)
408	48.801741	192.168.18.158	128.119.245.12	HTTP	660	GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1
415	49.034151	128.119.245.12	192.168.18.158	HTTP	544	HTTP/1.1 200 OK (text/html)

18 The server's response of initial HTTP GET message is 401 Unauthorized, which indicates that the request hasn't been applied because it lacks valid authentication credentials for the target resource.

Frame 52: 771 bytes on wire (6168 bits), 771 bytes captured (6168 bits) on interface \Device\NPF_{3ECB...}

Ethernet II, Src: 6a:e1:15:74:02:43 (6a:e1:15:74:02:43), Dst: AzureWaveTec_2a:81:41 (ec:2e:98:2a:81:41)

Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.18.158

Transmission Control Protocol, Src Port: 80, Dst Port: 50974, Seq: 1, Ack: 522, Len: 717

Hypertext Transfer Protocol

HTTP/1.1 401 Unauthorized\r\n

Response Version: HTTP/1.1

Status Code: 401

[Status Code Description: Unauthorized]

Response Phrase: Unauthorized

Date: Thu, 31 Oct 2024 20:00:06 GMT\r\n

Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.33 mod_perl/2.0.11 Perl/v5.16.3\r\n

WWW-Authenticate: Basic realm="wireshark-students only"\r\n

Content-Length: 381\r\n

Keep-Alive: timeout=5, max=100\r\n

Connection: Keep-Alive\r\n

Content-Type: text/html; charset=iso-8859-1\r\n

\r\n

[Request in frame: 5]

[Time since request: 0.267521000 seconds]

[Request URI: /wireshark-labs/protected_pages/HTTP-wireshark-file5.html]

0030 00 ed 01 26 00 00 48 54 54 50 2f 31 2e 31 20 34 ...& HT TP/1.1 401

0040 30 31 20 55 6e 61 75 74 68 6f 72 69 7a 65 64 0d 01 Unauth orized

0050 0a 44 61 74 65 3a 20 54 68 75 2c 20 33 31 20 4f Date: T hu, 31 O

0060 63 74 20 32 30 32 3a 20 32 30 3a 30 30 3a 30 36 ct 2024 20:00:06

0070 20 47 4d 54 0d 0a 53 65 72 76 65 72 3a 20 41 70 GMT- Se rver: Ap

0080 61 63 68 65 2f 32 2e 34 2e 36 20 28 43 65 6e 74 ache/2.4 .6 (Cent

0090 4f 53 29 20 4f 70 65 6e 53 53 4c 2f 31 2e 30 2e OS) Open SSL/1.0.

00a0 32 6b 2d 66 69 70 73 20 50 48 50 2f 37 2e 34 2e 2k-fips PHP/7.4.

00b0 33 33 20 6d 6f 64 5f 70 65 72 6c 2f 32 2e 30 2e 33 mod_p erl/2.0.

00c0 31 31 20 50 65 72 6c 2f 76 35 2e 31 36 2e 33 0d 11 Perl/ v5.16.3

00d0 0a 57 57 57 2d 41 75 74 68 65 6e 74 69 63 61 74 WW-Auth enticat

00e0 65 3a 20 42 61 73 69 63 20 72 65 61 6c 6d 3d 22 e: Basic realm="

00f0 77 69 72 65 73 68 61 72 6b 2d 73 74 75 64 65 6e wireshar k-studen

0100 74 73 20 6f 6e 6c 79 22 0d 0a 43 6f 6e 74 65 6e ts only" . Conen

0110 74 2d 4c 65 6e 67 74 68 3a 20 33 38 31 0d 0a 4b t-Length: 381. K

0120 65 65 70 2d 41 6c 69 76 65 3a 20 74 69 6d 65 6f eep-Aliv e: timeo

0130 75 74 3d 35 2c 20 6d 61 78 3d 31 30 30 0d 0a 43 ut=5, ma x=100 .C

0140 6f 6e 6e 65 63 74 69 6f 6e 3a 20 4b 65 65 70 2d onnection: Keep-

0150 41 6c 69 76 65 0d 0a 43 6f 6e 74 65 6e 74 2d 54 Alive .C ontent-T

0160 79 70 65 3a 20 74 65 78 74 2f 68 74 6d 6c 3b 20 ype: tex t/html;

0170 63 68 61 72 73 65 74 3d 69 73 6f 2d 38 38 35 39 charset= iso-8859

0180 2d 31 0d 0a 0d 0a 3c 21 44 4f 43 54 59 50 45 20 -1<!DOCTYPE

Wireshark - Packet 5 - Wi-Fi

Hypertext Transfer Protocol

GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1\r\n

Request Method: GET

Request URI: /wireshark-labs/protected_pages/HTTP-wireshark-file5.html

Request Version: HTTP/1.1

Host: gaia.cs.umass.edu\r\n

Connection: keep-alive\r\n

Upgrade-Insecure-Requests: 1\r\n

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/130.0.0.0 Safari/537.36\r\n

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.9\r\n

Accept-Encoding: gzip, deflate\r\n

Accept-Language: zh-TW,zh;q=0.9,en-US;q=0.8,en;q=0.7,zh-CN;q=0.6\r\n

[Response in frame: 52]

[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/protected_pages/HTTP-wireshark-file5.html]

6a e1 15 74 02 43 ec 2e 98 2a 81 41 08 00 45 00 j...t.C...*.A..E..

02 31 c0 ee 40 00 80 06 ef 0d c0 a8 12 9e 80 77 -1-@... ..w

f5 0c c7 1e 00 50 c2 f4 83 9b c8 35 f7 17 50 18P... ..S..P..

02 01 93 f3 00 00 47 45 54 20 2f 77 69 72 65 73GET /wires

68 61 72 6b 2d 6c 61 62 73 2f 70 72 6f 74 65 63 hark-lab s/prot

74 65 64 5f 70 61 67 65 73 2f 48 54 50 2d 77 ted_page s/HTTP-w

69 72 65 73 68 61 72 6b 2d 66 69 6c 65 35 2e 68 ireshark -file5.h

74 6d 6c 20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f tml HTTP /1.1-Ho

73 74 3a 20 67 61 69 61 2e 63 73 2e 75 6d 61 73 st: gaia .cs.umas

73 2e 65 64 75 0d 0a 43 6f 6e 6e 65 63 74 69 6f s.edu C onnectio

6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65 0d 0a 55 n: keep-alive-U

70 67 72 61 64 65 2d 49 6e 73 65 63 75 72 65 2d pgrade-I nsecure

52 65 71 75 65 73 74 73 3a 20 31 0d 0a 55 73 65 Requests : 1-Use

72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 r-Agent: Mozilla

2f 35 2e 30 28 57 69 6e 64 6f 77 73 20 4e 54 /5.0 (Wi ndows NT

20 31 30 2e 30 3b 20 57 69 6e 36 34 3b 20 78 36 10.0; W in64; x6

34 29 20 41 70 70 6c 65 57 65 62 4b 69 74 2f 35 4) Apple WebKit/5

33 37 2e 33 36 20 28 4b 48 54 4d 4c 2c 20 6c 69 37.36 (K HTML, li

Time: 3.193022 - Source: 192.168.18.158 - Destination: 128.119.245.12 - Protocol: HTTP - Length: 575 - Info: GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1

Show packet bytes Layout: Vertical (Stacked)

Wireshark - Packet 408 - Wi-Fi

Transmission Control Protocol, Src Port: 51050, Dst Port: 80, Seq: 1, Ack: 1, Len: 606

Hypertext Transfer Protocol

GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1\r\n

Request Method: GET

Request URI: /wireshark-labs/protected_pages/HTTP-wireshark-file5.html

Request Version: HTTP/1.1

Host: gaia.cs.umass.edu\r\n

Connection: keep-alive\r\n

Cache-Control: max-age=0\r\n

Authorization: Basic d2lyZXNoYXJrLXN0dWRlbnRzOm5ldHdvcmcs\r\n

Upgrade-Insecure-Requests: 1\r\n

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/130.0.0.0 Safari/537.36\r\n

Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.9\r\n

Accept-Encoding: gzip, deflate\r\n

Accept-Language: zh-TW,zh;q=0.9,en-US;q=0.8,en;q=0.7,zh-CN;q=0.6\r\n

78 2d 61 67 65 3d 30 0d 0a 41 75 74 68 6f 72 69 x-age=0. ..Authori

7a 61 74 69 6f 6e 3a 20 42 61 73 69 63 20 64 32 zation: Basic d2

6c 79 5a 58 4e 6f 59 58 4a 72 4c 58 4e 30 64 57 lyZXNoYXJrLXN0dW

52 6c 62 6e 52 7a 4f 6d 35 6c 64 48 64 76 63 6d RlbnRzOm5ldHdvcm

73 3d 0d 0a 55 70 67 72 61 64 65 2d 49 6e 73 65 s=..Upgr ade-Inse

63 75 72 65 2d 52 65 71 75 65 73 74 73 3a 20 31 cure-Req uests: 1

0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f ..User-A gent: Mo

7a 69 6c 6c 61 2f 35 2e 30 20 28 57 69 6e 64 6f zilla/5. 0 (Windo

77 73 20 4e 54 20 31 30 2e 30 3b 20 57 69 6e 36 ws NT 10 .0; Win6

34 3b 20 78 36 34 29 20 41 70 70 6c 65 57 65 62 4; x64) AppleWeb

4b 69 74 2f 35 33 37 2e 33 36 20 28 4b 48 54 4d Kit/537. 36 (KHTM

4c 2c 20 6c 69 6b 65 20 47 65 63 6b 6f 29 20 43 L, like Gecko) C

68 72 6f 6d 65 2f 31 33 30 2e 30 2e 30 2e 30 20 hrome/13 0.0.0.0

53 61 66 61 72 69 2f 35 33 37 2e 33 36 0d 0a 41 Safari/5 37.36..A

63 63 65 70 74 3a 20 74 65 78 74 2f 68 74 6d 6c ccept: t ext/html

2c 61 70 70 6c 69 63 61 74 69 6f 6e 2f 78 68 74 ,applica tion/xht

6d 6c 2b 78 6d 6c 2c 61 70 70 6c 69 63 61 74 69 ml+xml,a pplicati

Frame (660 bytes) Basic Credentials (26 bytes)

No.: 408 - Time: 48.801741 - Source: 192.168.18.158 - Destination: 128.119.245.12 - Protocol: HTTP - Length: 660 - Info: GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1

Show packet bytes Layout: Vertical (Stacked)